

FORM 2
THE PATENT ACT 1970
(39 of 1970)
&
The Patents Rules, 2003
COMPLETE SPECIFICATION
(See section 10 and M.13)

SELF-FLUSHING BEDPAN WITH HYGIENIC WASTE DISPOSAL SYSTEM FOR BEDRIDDEN CARE

Applicants(S) Name : New Horizon College of Engineering,
New Horizon Park,
Marathahalli outer ring road,
Bengaluru-560103.

Inventors(S) Name : Mrs. Shruthi G. R.
Assistant Professor,
Dept. of Information Science and Engineering,
New Horizon College of Engineering,
Marathahalli outer ring road,
Bengaluru-560103.

Mr. Shashikiran K. M.

Poornachandra Tejaswi
Department of Information Science and Engineering,
New Horizon College of Engineering
New Horizon Knowledge Park
Outer Ring Road, Near Marathalli
Bellandur (P), Bangalore-560103

Dr. D. Kalaivani
Associate Professor,
Dept. of Information Science and Engineering,
New Horizon College of Engineering,
Marathahalli outer ring road,
Bengaluru-560103.

Mrs. Priya N.
Assistant Professor,
Dept. of Information Science and Engineering,
New Horizon College of Engineering,
Marathahalli outer ring road,
Bengaluru-560103.

Mrs. Krishnaveni A.
Assistant Professor,
Dept. of Information Science and Engineering,
New Horizon College of Engineering,
Marathahalli outer ring road,

Bengaluru-560103.
Mrs. Rama Banshidar Dan
Assistant Professor,
Dept. of Information Science and Engineering,
New Horizon College of Engineering,
Marathahalli outer ring road,
Bengaluru-560103.

Mrs. Bibiana Jenifer J.
Assistant Professor,
Dept. of Information Science and Engineering,
New Horizon College of Engineering,
Marathahalli outer ring road,
Bengaluru-560103.

The following specification particularly describes the invention and the manner in which it is to be performed
SELF-FLUSHING BEDPAN WITH HYGIENIC WASTE DISPOSAL SYSTEM FOR BEDRIDDEN
CARE

Title: Self-Flushing Bedpan with Hygienic Waste Disposal System for Bedridden Care

Field of Invention

This invention relates to the field of medical devices, specifically bedpans used in hospitals, nursing homes, and home care settings. It is designed to improve the hygiene, comfort, and convenience of patients and caregivers by incorporating a water inlet for flushing and a faeces outlet for discharge.

Background

Traditional bedpans, used for collecting human waste, often require manual cleaning, which can be both cumbersome and unhygienic. This process increases the risk of contamination and poses a challenge for caregivers. To address these shortcomings, the present invention integrates a water inlet and a faeces outlet, offering a more hygienic and user-friendly solution.

Summary of the Invention

The bedpan described herein includes a body with an upper water inlet and a lower faeces outlet. This configuration allows for the introduction of water from the top to flush the interior of the bedpan, while the faeces can be discharged from the bottom through a secure outlet. The water inlet can be connected to an external water source, facilitating the rinsing process, and the faeces outlet is equipped with a closing mechanism to prevent leakage and maintain hygiene until the waste is ready to be disposed of.

KR1020200001333 - SMART PORTABLE BEDPAN FOR BEDRIDDEN PATIENT. The present invention relates to a smart portable bedpan for a bedridden patient, more specifically, to a smart portable bedpan for a bedridden patient capable of allowing faeces disposal of a bedridden patient who is difficult to walk and perform local cleaning of an anus (cleaning and drying), and additionally obtain user body information. The smart portable bedpan comprises a bedpan body, a faeces seat, a local cleaning device, and a physical constitution measuring module. The bedpan body supports a hip of a user to go to the toilet while sitting or lying on a bed. The faeces seat collects faeces discharged from a user body in a state of being mounted in the center of the bedpan body, and is received and discharged in a drawer type for convenience of collected faeces disposal. The local cleaning device is mounted inside the bedpan body, cleans and dries the anus and external genitalia of the user by an operation of a button operation, and removes odour of the faeces. The physical constitution measuring module obtains the user body information in a state of being mounted in a position of the bedpan body to which the user hip adheres.

US4011610 - Bedpan system. A complete bedpan system in which a mattress is aperture to accept a bedpan assembly, the bedpan being provided with a disposable liner to fully cover the surfaces engaged during usage and with the mouth of the liner being trapped between inter-engaging lips on the bedpan and on the mattress, hand-pocket recesses being provided to facilitate that trapping, with a sheet having a central aperture and a cuff to line and protect the aperture in the mattress, the cuff being trapped between the mattress and the bedpan, with a mattress plug to resiliently cover the bedpan when not in use, and provided with a rigid supporting base designed to center the plug with respect to the aperture in the mattress, and with an anti-contamination supporting plate underlying the mattress and the bedpan and serving to control the position of the bedpan relative to the springs and hence to control the position of the mattress relative to the springs.

This embodiment of the bedpan presents a novel and practical solution for managing waste in bedridden patients. Its design focuses on improving hygiene, reducing caregiver burden, and enhancing patient comfort. The integration of the water inlet, faeces outlet, and secure locking mechanisms demonstrates an innovative approach to modern bedpan design, particularly in healthcare settings where sanitation is critical. The Simple Water Inlet and Outlet System proposed in the present invention focuses on integrating a water inlet at the top for easy flushing and cleaning. This minimizes the need for complex electronic or automated cleaning systems. The absence of smart or power-operated components eliminates the reliance on electricity or battery-operated

devices, making it more accessible and functional in a wider range of care settings, including areas with limited power access.

Detailed Description

The bedpan (100) has a shallow, ergonomically designed body to comfortably fit under the patient. The material used is lightweight and easy to clean, typically plastic or stainless steel. The top surface includes a controlled opening (104) to ensure ease of use for the patient, while the sides are raised to prevent spillage. Located at the top of the bedpan, the water inlet (101) is designed to connect to a hose or other water source this ensures the bedpan (100) can be flushed after use, reducing the need for manual scrubbing. The inlet (101) directs water toward the center of the bedpan (100), washing away waste and debris. At the bottom of the bedpan, there is a faeces outlet (102) equipped with a cap or valve mechanism activated with the help of a button (103). This outlet (102) allows the contents of the bedpan to be easily disposed of without the need to lift or tilt the bedpan, minimizing the risk of spillage. The outlet (102) can be connected to a drain system or a collection container for hygienic disposal. Both the water inlet (101) and faeces outlet (102) are equipped with secure closing mechanisms (103 & 104) to ensure that the bedpan (100) remains sealed when not in use. The water inlet (101) has a one-way valve that prevents unintended water flow, while the faeces outlet (102) has a screw-on or sliding cap to prevent leakage. The integrated water flushing system reduces the need for intensive manual cleaning. After each use, water can be introduced from the inlet (101) to rinse the bedpan (100), and the waste is simultaneously discharged through the outlet (102). The bedpan (100) is designed to withstand multiple uses, and both the water inlet (101) and outlet (102) components are detachable for thorough cleaning. Low Maintenance by not integrating advanced cleaning or measuring modules, the bedpan (100) reduces the risk of mechanical failures and maintenance concerns. The design focuses on ease of cleaning and hygienic waste disposal, offering a practical solution to caregivers without needing specialized skills for upkeep. The inclusion of a water inlet (101) for flushing directly into the bedpan (100) and a faeces outlet (102) for disposal greatly enhances hygiene without the need for expensive and potentially failure-prone cleaning devices.

Brief Description of the Drawings

The embodiment of the figure 1 of the bedpan (100) has an ergonomically shaped surface, likely designed to fit comfortably beneath a patient's body. The water inlet (101) is located at the top of the bedpan (100), and it features a small circular one-way valve. This inlet (101) is used to introduce water into the bedpan (100) for flushing waste away after use. The one-way valve ensures that water can only flow into the bedpan (100), preventing any risk of contamination or backflow of waste or odours. The inlet (101) seems to have a one-way valve that is controlling the flow of water into the pan, rinsing away the waste. The faeces outlet (102) is located at the bottom section of the bedpan (100). This outlet (102) provides a method for disposing of waste without the need to physically remove or tip the bedpan (100). This part is designed to release waste into a connected drain or removable container for easier disposal. The top surface of the bedpan has a flat overlapping lid (105), which is collapsible with a button (104) separable for cleaning. This lid (105) covers the interior of the bedpan (100), likely designed to keep odours contained and facilitate hygienic handling when not in use. A button (103) to close the outlet (102), which indicates the presence of a locking mechanism (103) of the bedpan (100).

We Claim

1. A bedpan (100) comprising a body, a water inlet (101) positioned at the top of the body, and a faeces outlet (102) positioned at the bottom of the body, wherein the water inlet (101) is configured to be connected to a water source for flushing the bedpan (100) and the faeces outlet (102) is configured to discharge waste from the bedpan (100).
2. The bedpan (100) of claim 1, further comprising a one-way valve for the water inlet (101) that regulates the flow of water into the bedpan (100).
3. The bedpan (100) of claim 1, wherein the faeces outlet (102) is equipped with a cap or valve activated with the button (103) press to prevent leakage when the bedpan (100) is in use.
4. The bedpan (100) of claim 1, wherein the water inlet (101) and faeces outlet (102) are integrated in the bedpan (100) for ease of cleaning and maintenance.
5. The bedpan (100) of claim 1, wherein the bedpan (100) body is ergonomically shaped to comfortably fit under a patient and reduce the risk of spillage.
6. The bedpan (100) of claim 1, wherein the faeces outlet (102) is connected to a drain system or a collection container to facilitate hygienic disposal of waste.
7. The bedpan (100) of claim 1, wherein the body of the bedpan (100) is covered with a button (104) controlled collapsible lid (105) to avoid spillage and facilitate hygienic disposal of waste.
8. The bedpan (100) of claim 1, wherein the water inlet (101) has threads (106) to fit any kind of water source.
9. The bedpan (100) of claim 1, wherein the water inlet (101) one way valve (107) to prevent the water to exit from the inlet (101)

Date: 04/10/2024

**Dr. Manjunatha,
Principal, New Horizon College of Engineering**

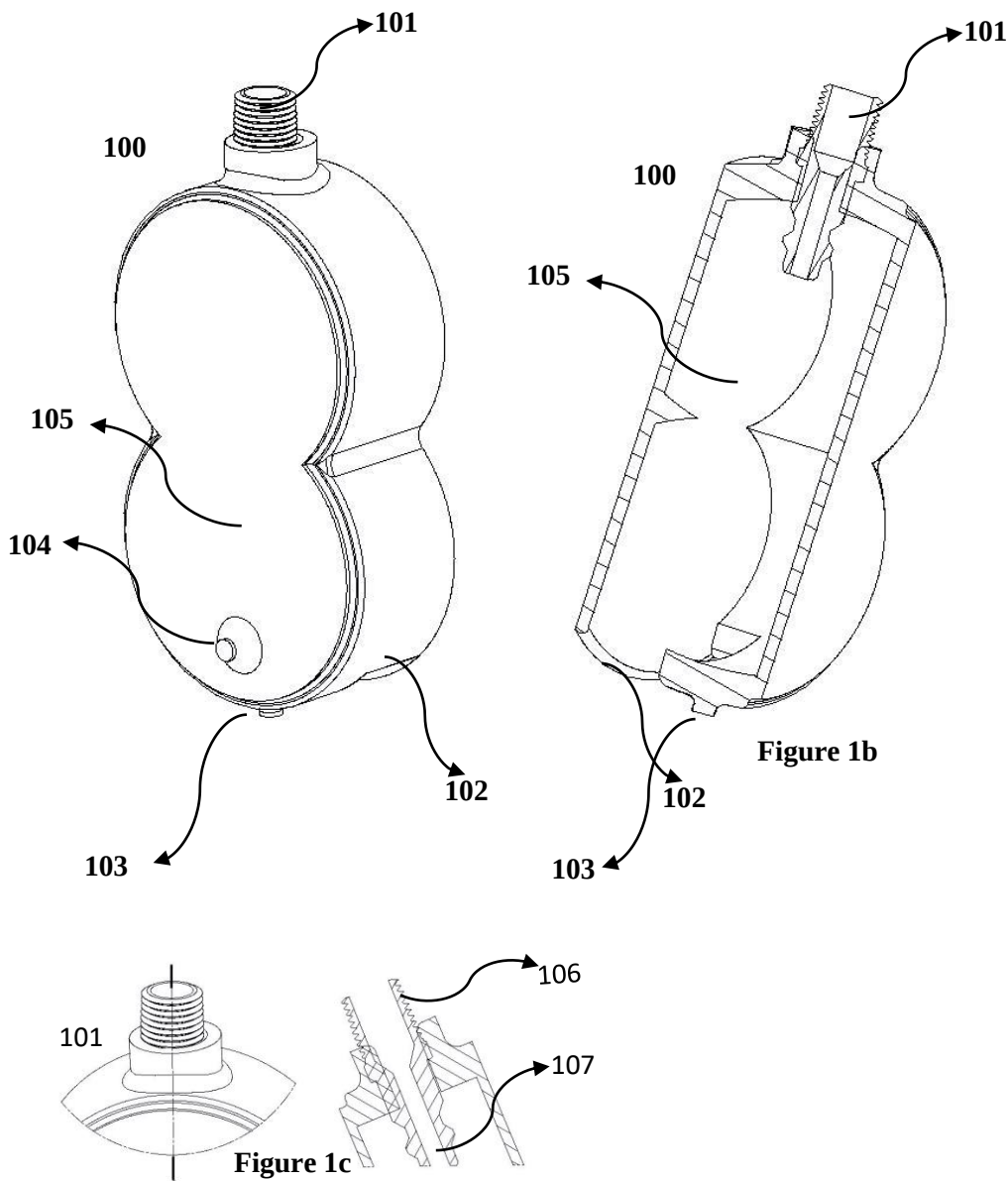


Figure 1

Abstract

The present invention relates to an ergonomically designed bedpan (100) featuring a one-way water inlet (101) and a faces outlet (102) for improved hygiene and ease of waste disposal, particularly suited for bedridden or limited-mobility patients. The bedpan (100) incorporates a circular one-way water inlet (101) valve at the top, allowing water to flow in for flushing waste without backflow, ensuring sanitary conditions. The bottom of the bedpan (100) includes a faces outlet (102), facilitating efficient waste disposal into a connected drain or removable container, eliminating the need for manual emptying. A collapsible lid (105) on the top surface of the bedpan (100) ensures odour containment and allows for easy cleaning when separated by pressing a button (104). Additionally, the outlet is controlled by a locking mechanism (103) that secures the bedpan (100) during and after use. This innovative design prioritizes patient comfort, hygiene, and caregiver convenience, providing a practical solution for waste management in healthcare and home settings.

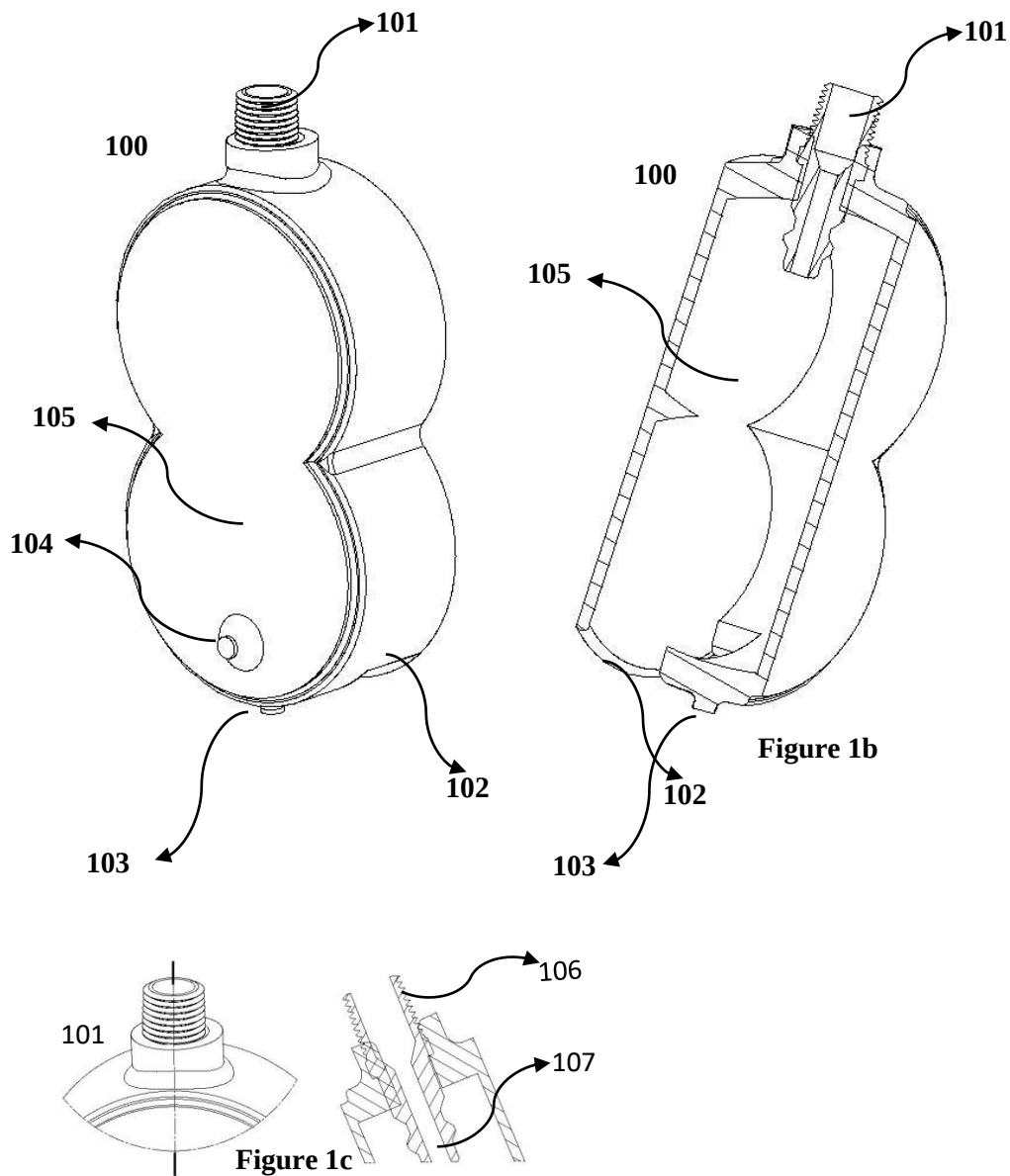


Figure 1